The University of the South Pacific

IS333 - Project Management

Assignment 1

Semester 1, 2026

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Question 1: Scheduling Techniques

a) Activity durations

The formula used to calculate the activity durations is,

T.E = (a + 4m + b)/6.

Activity label	Optimistic (a)	Most likely (m)	Pessimistic (b)	T.E = (a + 4m + b)/6 (Working)	Duration (Days)
A	3	4	7	= (3 + 4(4) + 7)/6 = 4.33	4
B	3	4	5	= (3 + 4(4) + 5)/6 = 4	4
C	8	10	12	= (8 + 4(10) + 12)/6 = 10	10
D	5	7	9	= (5 + 4(7) + 9)/6 = 7	7
E	7	8	11	= (7 + 4(8) + 11)/6 = 8.33	8
F	5	6	7	= (5 + 4(6) + 7)/6 = 6	6
G	3	4	8	= (3 + 4(4) + 8)/6 = 4.5	5
H	4	5	7	= (4 + 4(5) + 7)/6 = 5.17	5
I	10	12	14	= (10 + 4(12) + 14)/6 = 12	12
J	5	7	9	= (5 + 4(7) + 9)/6 = 7	7
K	7	8	11	= (7 + 4(8) + 11)/6 = 8.33	8
L	5	7	9	= (5 + 4(7) + 9)/6 = 7	7
M	10	14	16	= (10 + 4(14) + 16)/6 = 13.67	14
N	9	10	12	= (9 + 4(10) + 12)/6 = 10.17	10
O	7	8	11	= (7 + 4(8) + 11)/6 = 8.33	8
P	14	16	18	= (14 + 4(16) + 18)/6 = 16	16
Q	5	7	9	= (5 + 4(7) + 9)/6 = 7	7
R	10	13	16	= (10 + 4(13) + 16)/6 = 13	13
S	11	12	14	= (11 + 4(12) + 14)/6 = 12.17	12
T	18	20	22	= (18 + 4(20) + 22)/6 = 20	20
U	27	32	37	= (27 + 4(32) + 37)/6 = 32	32

V	6	7	10	= (6 + 4(7) + 10)/6 = 7.33	7
W	3	4	7	= (3 + 4(4) + 7)/6 = 4.33	4
X	7	8	11	= (7 + 4(8) + 11)/6 = 8.33	8

b) Activity table

The table below shows the activities along with its predecessors and durations after correcting the errors made and updated information provided in the specification document.

Activity label	Predecessors	Duration (Days)
A	-	4
B	A	4
C	B	10
D	C(SS3), 0(SS3)	7
E	D, F(SS4)	8
F	C	6
G	F(SS3), K(SS4)	5
H	-	5
I	H	12
J	B(SS5)	7
K	C, J	8
L	I(SS5)	7
M	J(SS2), L	14
N	F, M	10
O	-	8
P	M	16
Q	G, R(SS5)	7
R	N	13
S	E	12
T	P	20
U	T	32
V	S, W	7
W	E, G	4
X	Q, V	8

Some of the redundancies identified from the given AON diagram included:

i. B -> D
ii. N -> Q
iii. W -> X
iv. I -> J (as the new rule in the specification states that J would **only** start 5 days after B). Hence, additional constraint for J to start, which was I, is removed.

The predecessors were changed accordingly in the table above.

c) AON Diagram

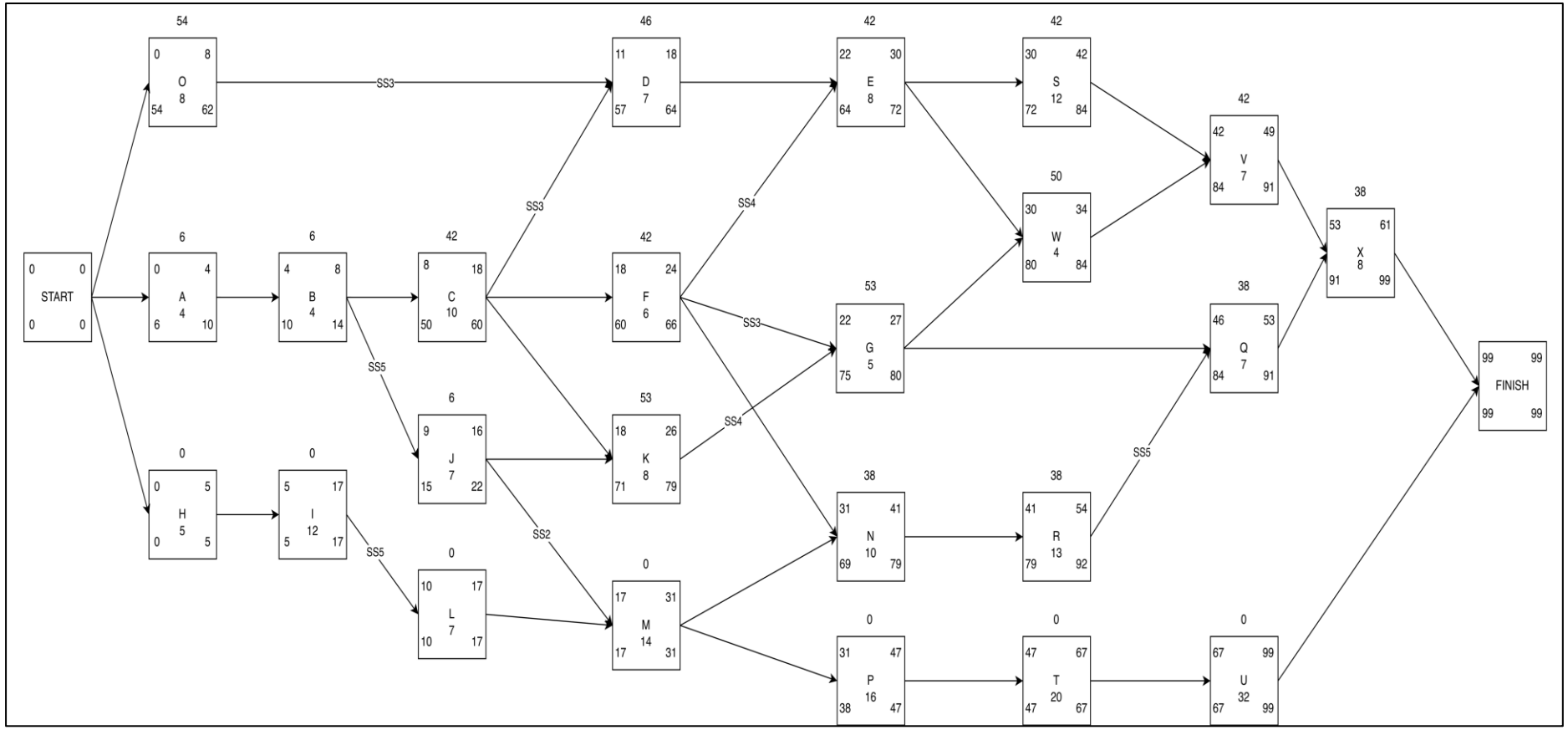
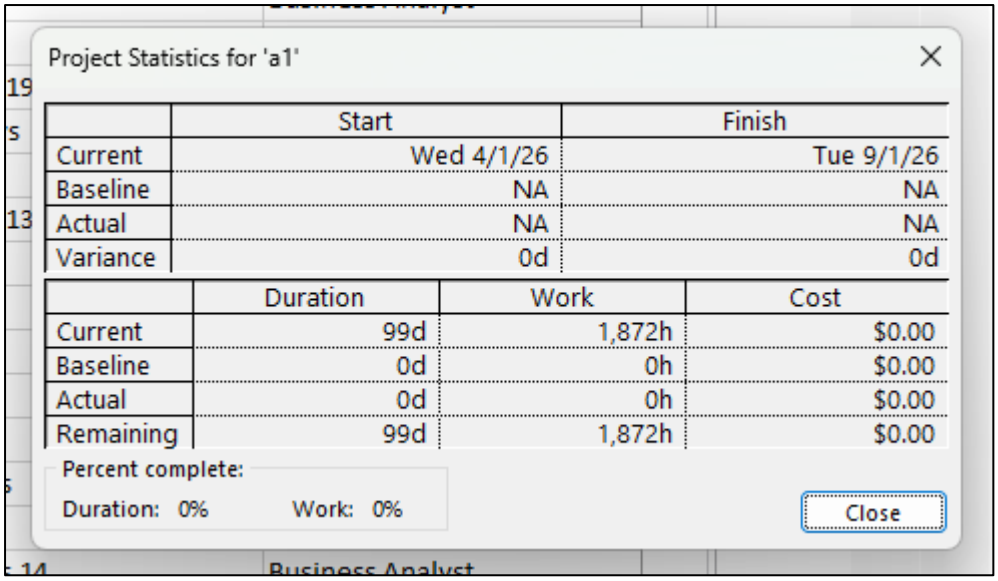


Figure 1, above, shows the completed AON Diagram together with ES, EF, LS & LF for each of the activities

d) Project Duration and Completion Date

- The duration of the project is **99 working days**.
- The completion date of the project is the **1st of September, 2026**. **(1/9/26)**.



Project Statistics for 'a1'

	Start	Finish
Current	Wed 4/1/26	Tue 9/1/26
Baseline	NA	NA
Actual	NA	NA
Variance	0d	0d

	Duration	Work	Cost
Current	99d	1,872h	$0.00
Baseline	0d	0h	$0.00
Actual	0d	0h	$0.00
Remaining	99d	1,872h	$0.00

Percent complete:

Duration: 0% Work: 0%

Close

Figure 2, obtained from MS Project, shows the Project Statistics such as duration and completion date.

e) Distinct Paths along with Durations

Path	Working	Duration
O->D->E->S->V->X	3(ss3) + 7 + 8 + 12 + 7 + 8	45
O->D->E->W->V->X	3(ss3) + 7 + 8 + 4 + 7 + 8	37
H->I->L->M->N->R->Q->X	5 + 5(ss5) + 7 + 14 + 10 + 13 + 8	62
H->I->L->M->P->T->U	5 + 5(ss5) + 7 + 14 + 16 + 20 + 32	99 (Critical Path)
A->B->J->M->N->R->Q->X	4 + 5(ss5) + 2(ss2) + 14 + 10 + 13 + 8	56
A->B->J->M->P->T->U	4 + 5(ss5) + 2(ss2) + 14 + 16 + 20 + 32	93
A->B->J->K->G->W->V->X	4 + 5(ss5) + 7 + 4(ss4) + 5 + 4 + 7 + 8	44
A->B->J->K->G->Q->X	4 + 5(ss5) + 7 + 4(ss4) + 5 + 7 + 8	40
A->B->C->K->G->Q->X	4 + 4 + 10 + 4(ss4) + 5 + 7 + 8	42
A->B->C->K->G->W->V->X	4 + 4 + 10 + 4(ss4) + 5 + 4 + 7 + 8	46
A->B->C->F->N->R->Q->X	4 + 4 + 10 + 6 + 10 + 13 + 8	55
A->B->C->F->G->W->V->X	4 + 4 + 10 + 3(ss3) + 5 + 4 + 7 + 8	45
A->B->C->F->G->Q->X	4 + 4 + 10 + 3(ss3) + 5 + 7 + 8	41
A->B->C->F->E->S->V->X	4 + 4 + 10 + 4(ss4) + 8 + 12 + 7 + 8	57
A->B->C->F->E->W->V->X	4 + 4 + 10 + 4(ss4) + 8 + 4 + 7 + 8	49
A->B->C->D->E->S->V->X	4 + 4 + 3(ss3) + 7 + 8 + 12 + 7 + 8	53
A->B->C->D->E->W->V->X	4 + 4 + 3(ss3) + 7 + 8 + 4 + 7 + 8	45

f) Gantt Chart

MS Project file is also uploaded to the Dropbox.

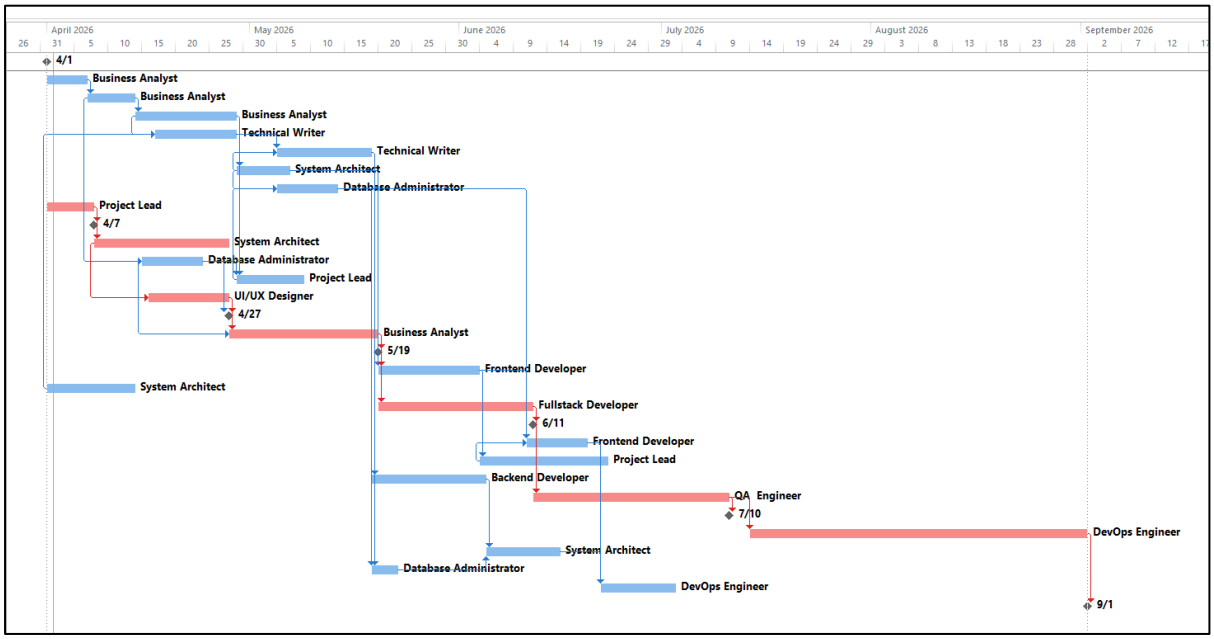


Figure 3, obtained from MS Project, shows the Gantt Chart of the Project.

g) One Issue Weighing in Favor of and Against Project Crashing

The advantage of project crashing, as Pinto (2019, p. 360) mentioned, is that more resources will be allocated to the project or critical activities to reduce delays, maintain the schedule or accelerate the project, thus the resource conflicts can be addressed. In the Ringinrejo road project, through the successful crashing of the critical excavation activity, the team was able to compress the remaining duration from 273 days to 184 days despite being 89 days behind schedule, thus realigning to the schedule (Dewi & Kamandang, 2023).

Additionally, as stated by Polat et al. (2020), with project crashing the ability to meet or beat contract deadlines could unlock early finish bonuses, thus becoming a motivator for contractors to purse project crashing. However, the success of project crashing is dependent on the level of resource constraint the project is operating under. This creates potential risks that can compromise project quality in the process (Pinto, 2019, p. 359). In the Ringinrejo road project, the cheapest available crashing scenario depleted over 85% of the remaining project budget (Dewi & Kamandang, 2023). Moreover, Turkoglu et al. (2020) reinforces this by stating that aggressive crashing could cause financial and quality risks that extend beyond the immediate cost of an activity's crash duration, harming the flexibility of future disruptions.

h) Two Important Questions to Consider Before Deciding whether to Crash a Project or Not

Two questions that should be taken into consideration before crashing a project are firstly, **what are the direct costs of accelerating an activity?** And secondly, **what wider consequences might result from project activities crashing?** The options should be evaluated by comparing the crash cost with the amount of time saving, identifying activities on the critical path, and choosing the option with the best time–cost trade-off (Pinto, 2019, p. 363).

For the first question, a time-cost trade-off must be determined with calculation of each critical path activity's cost slope and compression levels where it is optimal to do crashing without outweighing cost savings (Turkoglu et al., 2020). Adding to this, Dewi & Kamandang (2023) demonstrated through the evaluation of four distinct equipment-addition scenarios that time savings is not the monolithic optimal solution but rather the scenario that carried the lowest cost slope while also being able to stay within the schedule with the remaining budget.

In terms of the second question, contract clauses governing penalties and early finish bonuses alter the optimal crashing decision with changes to completion durations and which activities should be crashed first (Polat et al., 2020). Therefore, not following these clauses as a project manager results in suboptimal crashing. In addition, as seen in the SDN Kutisari 2 project (Firdaus et al., 2023), the order of crashing activities is necessary with certain activities vulnerable to quality degradation if accelerated without proper scheduling.

References

Pinto, J. K. (2019). Project management: Achieving competitive advantage (5th ed.). Pearson. https://www.pearson.com/en-us/subject-catalog/p/project-management-achieving-competitive-advantage/P200000006022/9780137477258

Firdaus, M. B., 1, Tjendani, H. T., Witjaksana, B., & University 17 August 1945 Surabaya. (2023). ACCELERATION ANALYSIS USING THE CRASHING METHOD IN THE SDN KUTISARI 2 DEVELOPMENT PROJECT. In INTERNATIONAL JOURNAL ON ADVANCED TECHNOLOGY, ENGINEERING, AND INFORMATION SYSTEM (Vol. 2, Issue 4, pp. 366–367) [Journal-article]. https://pdfs.semanticscholar.org/c2f9/3c68751cd9948e3d2f95ef5169258ff13035.pdf

Dewi, P. F., & Kamandang, Z. R. (2023). Optimizing project performance by applying the crashing method to road construction project. *Advance Sustainable Science Engineering and Technology*, *5*(2), 0230203. https://doi.org/10.26877/asset.v5i2.15944

Turkoglu, H., Polat, G., Damci, A., & Akin, F. D. (2020). Project Crashing with Crash Duration Consumption Rate: An Illustrative Example. In Proceedings of the Creative Construction e-Conference (p. 046). https://www.researchgate.net/profile/Firat-Akin/publication/342918114_Project_Crashing_with_Crash_Duration_Consumption_Rate_An_Illustrative_Example/links/5f0d6fbb92851cc4bf72fe26/Project-Crashing-with-Crash-Duration-Consumption-Rate-An-Illustrative-Example.pdf

Polat, G., Akin, F. D., Turkoglu, H., & Damci, A. (2020). A Project Crashing Strategy Considering Contract Clauses and Quality Considerations: An Illustrative Example. In Proceedings of the Creative Construction e-Conference (p. 040). https://repozitorium.omikk.bme.hu/server/api/core/bitstreams/448a6d48-31f3-4178-a399-71f2cc2cfbae/content

Appendix

Terms of Reference (ToR)

1.0 Group Members

Name	Student ID	Student Email
Anav Chand	S11221203	S11221203@student.usp.ac.fj
Vishant Kumar	S11230430	S11230430@student.usp.ac.fj
Zhixian Chen	S11230686	S11230686@student.usp.ac.fj
Rohan Nandan	S11234883	S11234883@student.usp.ac.fj

2.0 Team Purpose

This team of four is formed to complete the IS333 Assignment 1 of Semester 1, 2026. The team is committed to analyzing the Truemart Mall project of Truemart LTD collaboratively and applying the project management scheduling techniques learned as part of the course in order to address the required tasks in this assignment. To ensure that a high-quality and well-structured assignment is submitted on time, the team will be dedicated to ensuring seamless and professional communication, utilizing professional knowledge effectively and distributing tasks fairly to guarantee that each member participates and contributes to the assignment equally.

3.0 Team Roles

Member Name	Role
Vishant Kumar	Team Leader
Anav Chand	Scheduler, Analyzer
Rohan Nandan	Scheduler, Analyzer
Zhixian Chen	Scheduler, Analyzer

4.0 Skill Inventory

Member Name	Skill
Vishant Kumar	• Leadership and coordination • Team tasks organization • Critical analysis and mathematical skills • Diagram Drawing
Anav Chand	• Critical analysis and mathematical skills • Integration and Inspection capabilities • Special attention to technical details • Diagram Drawing
Rohan Nandan	• Critical analysis and mathematical skills • Project coordination and planning skills • Structural thinking and analyzing • Diagram Drawing
Zhixian Chen	• Critical analysis and mathematical skills • Diagram Drawing • Information Collecting

5.0 Resources

i. **Assignment 1 Specification**

- The instructions for the assignment. It provides task requirements, deadlines, and marking criteria for this assignment.

ii. **The Internet and USP Library**

- The team will utilize the resources carefully to clarify concepts of project management and required reports format to ensure compliance with academic standards.

iii. **Lecture/Lab Notes and Published Course Materials**

- These will be used as guidance to apply the required techniques to solve tasks of this assignment. These materials also provide examples and practice relevant to AON diagrams, forward pass, backward pass, float, and Gantt chart construction.

iv. **Drawing and Scheduling Software**

- The team will use appropriate tools for this assignment, such as MS Visio, MS Project, Draw.io, or other approved tools learned in class.

6.0 Governance

All decisions regarding this assignment will be thoroughly discussed. The final decision will take into account the opinions of each member to maximize the decision's rationality and fair member participation. Governance aims to ensure fairness and accountability throughout the collaboration, thereby achieving the overall team goal: a high-quality and well-structured assignment is submitted on time.

7.0 Codes of Conduct and Ethics

To ensure effective team collaboration and excellent quality of the final report, all team members agree to obey the following code of conduct. The following rules are designed to maintain a professional and respectful environment:

- **Communicate openly and honestly**: Team members are free to express their true opinions about current progress and team status without being criticized.
- **Actively participate**: all team members are expected to attend meetings and discussions to synchronize work progress and exchange ideas in order to improve overall quality of work.
- Resolving issues as a team: addressing conflicts or problems collaboratively, team members must provide assistance for each other whenever possible.
- **Complete the work within deadlines:** all team members are expected to complete their work on time
- Reviewing all work before submission: a thorough inspection at a later stage of a project is essential for overall quality.
- **Maintain confidentiality**: team members must not copy from other's work without acknowledgement or allow others to use teamwork without consent.

8.0 Decision Making Policy

- **Consensus Driven**: Each decision will be made through agreement after discussion.
- Timeliness: Decisions are made within a dedicated time schedule to maintain the project process.
- **Flexibility**: To ensure adaptability throughout the development process, the team will re-evaluate and adjust decisions when new information becomes available or project circumstances change
- **Transparency**: Each member of the team will be able to stay informed about the decision-making process and the content of decisions through meeting records and group chat, whether they were present in the meeting or not.
- **Accountability**: Once a decision is finalized, each group member is responsible for adhering to and executing it.

IS333 Project Management Semester 1, 2026
Assignment 1 Mark Allocation Agreement

After having discussed as a team, we recommend the following mark allocation to each team member based on contribution or lack of it throughout the assignment.

Member ID	Percentage of Assignment 1 marks	Task Completed
S11230430	100%	a, b & c
S11221203	100%	c, d & f
S11234883	100%	c, e, & h
S11230686	100%	c, g & h

Signed

Member name	ID	Signature
Vishant Kumar	S11230430	*vishant*
Anav Chand	S11221203	*anav*
Rohan Nandan	S11234883	*rohan*
Zhixian Chen	S11230686	*zhixian*